Therefore, since this should hold for an arbitrary domain $\mathcal{A}$, we must have

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\rho \boldsymbol{f}).$$

This is called the Liouville equation or continuity equation. It tells us how the distribution of phase points varies in time and space.

An alternative form for the continuity equation is:

$$\frac{\partial \rho}{\partial t} = -\sum_i \left[\rho \frac{\partial f_i}{\partial z_i} + f_i \frac{\partial \rho}{\partial z_i} \right].$$

The operator $\mathcal{M}_f$, defined for a scalar valued function w by $\mathcal{M}_f w = -\nabla \cdot (w\boldsymbol{f})$, is called the Liouvillian operator.

There is a close relationship between $\mathcal{M}_f$ and the Lie derivative discussed earlier. Let u, v be two scalar valued functions in $\mathbb{R}^m$ which tend to zero sufficiently rapidly at infinity. We truncate the domain to a large box $\mathcal{A}_\ell$ with edge of length 2ℓ centered at the origin, approximating the inner product by:

$$\langle u, \mathcal{L}_f v \rangle \approx \int_{\mathcal{A}_\ell} u \mathcal{L}_f v \mathrm{d}\omega = \int_{\mathcal{A}_\ell} u \sum_i f_i \frac{\partial v}{\partial z_i} \mathrm{d}\omega.$$

Splitting this sum up, we treat each term by integration by parts

$$\int_{\mathcal{A}_\ell} u \sum_i f_i \frac{\partial v}{\partial z_i} \mathrm{d}\omega = \sum_i \int_{\mathcal{A}_\ell} u f_i \frac{\partial v}{\partial z_i} \mathrm{d}\omega \tag{5.4}$$

$$= B - \sum_i \int_{\mathcal{A}_\ell} v \frac{\partial}{\partial z_i} (u f_i) \mathrm{d}\omega, \tag{5.5}$$

where the term $B = \sum_i \int_{\partial \mathcal{A}_\ell} u f_i v \nu_i \mathrm{d}\sigma$, is a surface integral over the boundary of $\mathcal{A}_\ell$, where ν_i is the i'th component of the unit outward normal to the surface $\partial \mathcal{A}_\ell$ bounding the region $\mathcal{A}_\ell$. By our assumptions that u and v are rapidly decaying we know that $B \to 0$, as $\ell \to \infty$. Taking the limit, we are thus left with the equation

$$\langle u, \mathcal{L}_f v \rangle = -\sum_i \int_{\mathbb{R}^m} v \frac{\partial}{\partial z_i} (u f_i) \mathrm{d}\omega = \langle \mathcal{M}_f u, v \rangle.$$

We refer to $\mathcal{M}_f$ as the *adjoint* operator[3] of $\mathcal{L}_f$ and we write $\mathcal{M}_f = \mathcal{L}_f^\dagger$, where

$$\langle u, \mathcal{L}_f v\rangle = \langle \mathcal{L}_f^\dagger u, v\rangle$$

and

$$\mathcal{L}_f^\dagger u = -\nabla \cdot (u\boldsymbol{f}).$$

The density ρ evolves as the solution of the partial differential equation

$$\frac{\partial \rho}{\partial t} = \mathcal{L}_f^\dagger \rho,$$

where we interpret ρ as a function of both $\boldsymbol{z}$ and t and view $\mathcal{L}_f^\dagger$ as only operating on the spatial components. This means, formally, $\rho(\boldsymbol{z}, t) = e^{t\mathcal{L}_f^\dagger}\rho(\boldsymbol{z}, 0)$. We can think of $\exp(t\mathcal{L}_f^\dagger)$ as defining the *propagator* of measure.

5.2.3 The Liouville Equation for Hamiltonian Systems

Until now we have considered a general system $\dot{\boldsymbol{z}} = \boldsymbol{f}(\boldsymbol{z})$. What happens when $\boldsymbol{f}$ is derived from a Hamiltonian?

From the above we know that for a smooth scalar function $u = u(\boldsymbol{z})$,

$$\mathcal{L}_f^\dagger u = -\nabla \cdot (u\boldsymbol{f}) = -\boldsymbol{f} \cdot \nabla u - u\nabla \cdot \boldsymbol{f}.$$

The term $\nabla \cdot \boldsymbol{f}$ vanishes for $\boldsymbol{f} = \boldsymbol{J}\nabla H$. Thus we have

$$\mathcal{L}_f^\dagger u = -\boldsymbol{f} \cdot \nabla u = -\mathcal{L}_f u.$$

If this holds for all smooth u, we write

$$\mathcal{L}_f^\dagger = -\mathcal{L}_f.$$

[3]The adjoint operator (acting on functions) is not to be confused with the adjoint of a map (which acts on points in phase space) encountered in Chap. 2.